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Simon Says!

A colour-sequence memory game built with MIT App Inventor



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Diploma in Computer Science · UiTM Jasin

Course: ISP250 — Introduction to Software Programming

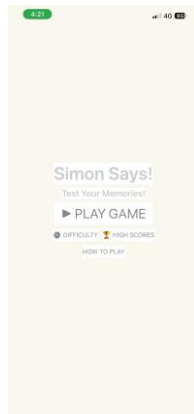
1. PROJECT OVERVIEW

Simon Says! is a fully functional mobile memory game developed for Android using MIT App Inventor, a block-based visual programming environment. Inspired by the classic Simon electronic game, players must memorise and repeat a growing sequence of colour flashes. The application was built as the final project for ISP250 (Introduction to Software Programming) during Semester 4 at UiTM Jasin.

Platform	Android (MIT App Inventor)
Tool	MIT App Inventor 2 (Block-based programming)
Number of Screens	6 (MainMenu, GameScreen, GameOverScreen, DifficultyScreen, HighScoresScreen, InstructionsScreen)
Data Storage	TinyDB (on-device persistent storage)
Course	ISP250 — Introduction to Software Programming, Semester 4, 2024

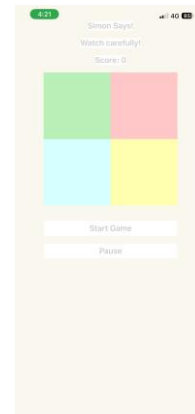
2. APP SCREENS

The application consists of six screens, each serving a distinct purpose in the user journey.



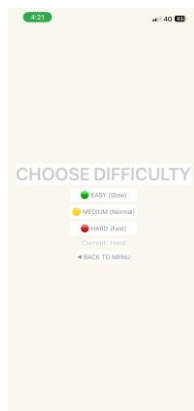
Main menu

Navigation hub with Play, Difficulty, High Scores and How to Play buttons. Background music plays and resumes automatically on return.



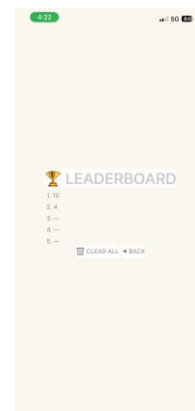
Gameplay

Four colour tiles flash in sequence. Player taps them in order. Score increments each round.



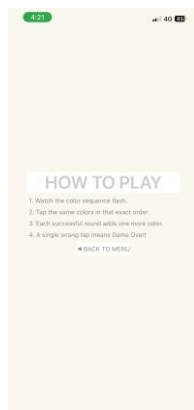
Instructions

Step-by-step rules for first-time players. Back to Menu button to return.



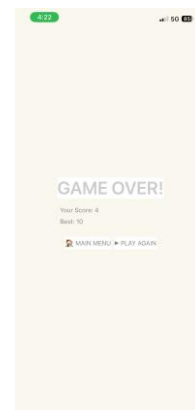
Difficulty

Three speed settings: Easy (800 ms), Medium (600 ms), Hard (350 ms). Current selection shown live.



Leaderboard

Top 5 scores stored in TinyDB. Clear All button with confirmation dialog.



Game over

Displays score and all-time best. New record flagged. Play Again or Main Menu options.

3. KEY FEATURES

Six core systems were designed and implemented from scratch.

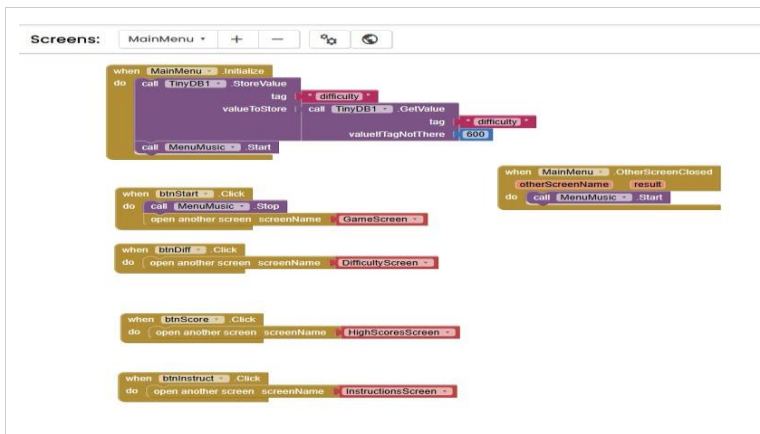
3-level difficulty Flash speed stored in TinyDB as a timer interval: Easy 800 ms, Medium 600 ms, Hard 350 ms. Persists across app restarts.	Persistent leaderboard Top 5 scores sorted using an insertion sort algorithm and stored in TinyDB. New all-time bests detected and flagged.
Sound & music Background music on menu and game screens. A beep plays on each colour flash. A buzz plays on wrong input. Input validation isFlashing flag blocks colour button taps during the sequence display phase, preventing false or early inputs.	Pause & resume Full pause overlay with Resume and Menu buttons. Both Clock timers and music halt and restore cleanly. Instant replay Play Again sends a 'restart' flag back to GameScreen via OtherScreenClosed, resetting state without the menu.

4. SKILLS DEMONSTRATED

- TinyDB persistence — saving and reading difficulty settings and high score lists across sessions
- Clock-based animation — dual-clock system (Clock1 for flash timing, Clock2 for colour reset)
- Insertion sort algorithm — sorting new scores into the stored top-5 list in GameOverScreen
- Multi-screen navigation — open/close screen with value passing (result: restart / menu)
- Audio management — playing, stopping, and pausing background music and sound effects
- Global state variables — sequence, score, playerId, flashIndex, isFlashing, difficulty
- List manipulation — add items, select item, length of list, create empty list
- Event-driven programming — button click, screen initialize, clock timer, other screen closed

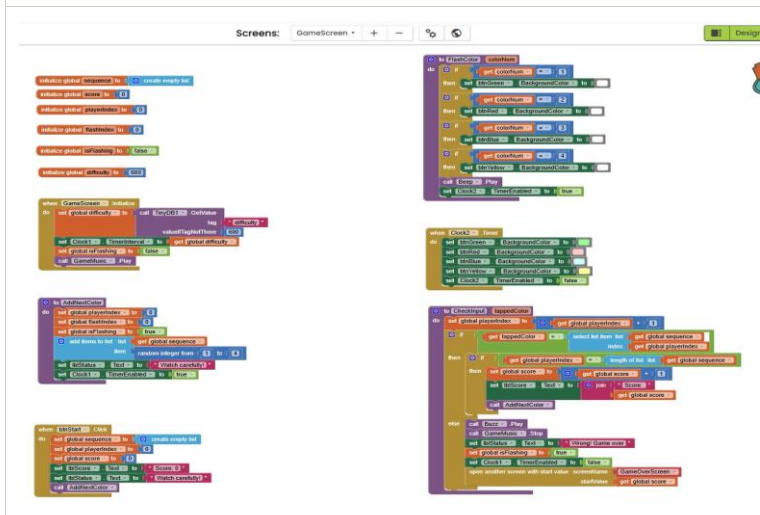
5. BLOCK CODE — SCREEN BY SCREEN

All application logic is implemented using MIT App Inventor's visual block editor. Below are the block diagrams for each screen.



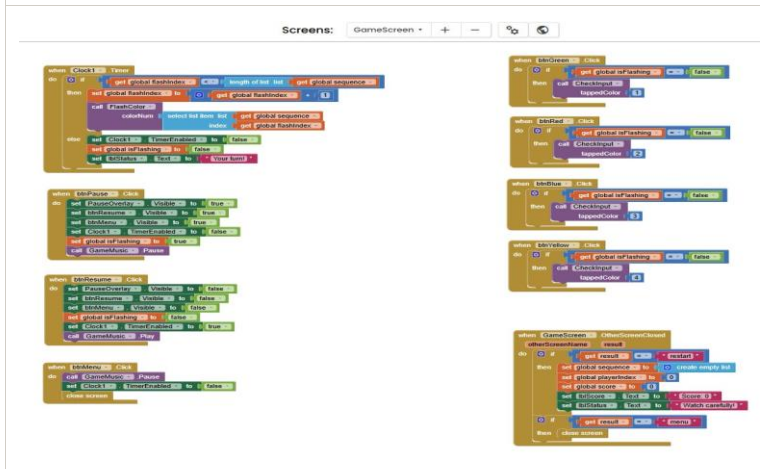
MainMenu

Initialises difficulty from TinyDB on startup. Starts background music. Navigates to GameScreen, DifficultyScreen, HighScoresScreen, and InstructionsScreen. Resumes music via OtherScreenClosed.



GameScreen (1 of 2)

Global variables: sequence, score, playerIndex, flashIndex, isFlashing, difficulty. AddNextColor, CheckInput, and FlashColor procedures. Start button handler resets all state and calls AddNextColor.



GameScreen (2 of 2)

Clock1 timer advances flashIndex to animate the sequence. Clock2 timer resets colours after each flash. Four colour button handlers guarded by isFlashing. Pause, Resume, and Menu button handlers.

DifficultyScreen

DisplayActiveDifficulty reads TinyDB and updates the label. Easy/Medium/Hard buttons store 800/600/350 ms and immediately refresh the label. Back closes the screen.

HighScoresScreen

RefreshLeaderboard reads `highscore_list` from TinyDB and fills five score labels. Clear button shows a notifier dialog asking for confirmation before wiping both TinyDB tags.

GameOverScreen

On init: plays game-over sound, shows score, runs insertion-sort to place score in top-5 list, detects new record. Play Again closes with 'restart' flag; Menu closes with 'menu' flag.

6 . A B O U T T H E D E V E L O P E R

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Programme

Diploma in Computer Science

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Course

ISP250 — Introduction to Software Programming

Semester

Semester 4

Year

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